Practice Questions for 2024-09-19-DualityExamples

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1 Questions

1. In the Diet Problem (Slides 2-8), the primal problem minimizes the cost of food while meeting nutritional requirements. The dual problem, introduced in Slides 7 and 8, maximizes the profit from selling nutrient pills. Explain the economic interpretation of the dual variable y_i in the context of the dual problem's objective function, $\sum_i b_i y_i$.

   a. y_i represents the amount of nutrient i consumed.

   b. y_i represents the cost of food j containing nutrient i.

   c. y_i represents the maximum price per unit of nutrient i such that selling pills is more profitable than selling food.

   d. y_i represents the minimum amount of nutrient i required in the diet.

   e. y_i represents the profit obtained from selling one unit of food j containing nutrient i.

2. Consider the Diet Problem's primal formulation (Slide 5): minimize $\sum_j c_j x_j$ subject to $\sum_j a_{ij} x_j \geq b_i$ and $x_j \geq 0$. The dual problem (Slide 8) is formulated as: maximize $\sum_i b_i y_i$ subject to $\sum_i y_i a_{ij} \leq c_j$ and $y_i \geq 0$. Explain the economic interpretation of the dual constraint $\sum_i y_i a_{ij} \leq c_j$.

   a. The total cost of nutrients in food j must exceed the price of food j.

   b. The total price of pills to replace the nutrients in food j must be less than or equal to the price of food j.

   c. The minimum intake of nutrient i must be met by the consumption of food j.

   d. The profit from selling pills for nutrient i must be greater than the cost of food j.

   e. The amount of nutrient i in food j must be non-negative.

3. Weak duality states that for any feasible solution x to the primal problem and any feasible solution y to the dual problem, the objective function value of the primal problem is less than or equal to the objective function value of the dual problem. In the context of the Diet Problem (Slides 2-8), explain why this inequality holds intuitively.

   a. The primal problem minimizes cost, while the dual problem maximizes profit, so the minimum cost will always be less than the maximum profit.

   b. The primal problem uses food, while the dual problem uses pills, so the costs are inherently different.

   c. The dual problem provides an upper bound on the optimal cost of the primal problem, representing the maximum profit that could be made by selling nutrient pills.

d. The primal problem's constraints are inequalities, while the dual problem's constraints are equalities, leading to different objective function values.

e. The primal and dual problems have different objective functions, so there is no direct relationship between their values.

4. In the Diet Problem (Slides 2-8), strong duality states that if both the primal and dual problems have optimal solutions, then the optimal objective function values are equal. Explain the economic interpretation of this equality in the context of the problem.

   a. The minimum cost of the diet is always equal to the maximum profit from selling pills, regardless of the specific food and pill prices.

   b. There is no economic interpretation; it's purely a mathematical result.

   c. If the minimum cost diet is found, then there exists a set of pill prices such that the maximum profit from selling those pills exactly equals the minimum cost of the diet.

   d. The equality only holds if the pill prices are equal to the food prices.

   e. The equality only holds if the minimum nutrient intake is zero.

5. Consider the primal Diet Problem (Slide 5): minimize $\sum_j c_j x_j$ subject to $\sum_j a_{ij} x_j \geq b_i$ and $x_j \geq 0$. Its dual (Slide 8) is: maximize $\sum_i b_i y_i$ subject to $\sum_i y_i a_{ij} \leq c_j$ and $y_i \geq 0$. If the primal problem represents minimizing the cost of a diet meeting nutritional requirements, what does the dual problem represent economically, and what is the economic interpretation of the dual variable y_i?

   a. The dual problem represents maximizing the profit from selling individual nutrients, and y_i represents the profit margin per unit of nutrient i.

   b. The dual problem represents minimizing the cost of producing nutrient pills, and y_i represents the production cost per unit of nutrient i.

   c. The dual problem represents maximizing the profit from selling a complete set of nutrient pills, and y_i represents the price per unit of a pill containing nutrient i.

   d. The dual problem represents minimizing the cost of a diet while maximizing nutritional value, and y_i represents the relative importance of nutrient i.

   e. The dual problem represents maximizing the revenue from selling food, and y_i represents the price per unit of nutrient i in the food.

6. In the Diet Problem (Slides 2-8), the primal problem minimizes the cost of food while meeting nutritional requirements. The dual problem (Slides 7-8) maximizes the profit from selling nutrient pills. If the optimal solution to the primal problem is x^*, resulting in a minimum cost of $C^* = \sum_j c_j x_j^*$, and the optimal solution to the dual problem is y^*, resulting in a maximum profit of $P^* = \sum_i b_i y_i^*$, what relationship exists between C^* and P^* according to strong duality, and what is the economic interpretation of this relationship?

   a. $C^* < P^*$; The profit from selling pills is always greater than the cost of the optimal diet.

   b. $C^* > P^*$; The cost of the optimal diet is always greater than the profit from selling pills.

   c. $C^* = P^*$; The minimum cost of the diet equals the maximum profit from selling pills.

   d. $C^* \leq P^*$; The minimum cost of the diet is less than or equal to the maximum profit from selling pills.

e. There is no guaranteed relationship between C^* and P^*.

7. Consider the Diet Problem's primal formulation (Slide 5): minimize $\sum_j c_j x_j$ subject to $\sum_j a_{ij} x_j \geq b_i$ and $x_j \geq 0$. The dual problem (Slide 8) is: maximize $\sum_i b_i y_i$ subject to $\sum_i y_i a_{ij} \leq c_j$ and $y_i \geq 0$. Provide an economic interpretation of the dual constraint $\sum_i y_i a_{ij} \leq c_j$ in the context of the problem.

   a. The total amount of nutrient i consumed must be greater than or equal to the minimum requirement b_i.

   b. The price of food j must be less than or equal to the total value of its nutrients when sold as pills.

   c. The minimum intake of nutrient i must be met by the consumption of foods.

   d. The total cost of the diet must be minimized.

   e. The unit price of each nutrient pill must be non-negative.

8. In the context of the Diet Problem (Slides 2-8), explain the economic interpretation of the dual variable y_i in the dual problem's objective function, $\sum_i b_i y_i$.

   a. y_i represents the amount of food j consumed.

   b. y_i represents the minimum required intake of nutrient i.

   c. y_i represents the unit price of a pill providing nutrient i.

   d. y_i represents the amount of nutrient i in food j.

   e. y_i represents the cost of food j.

9. Weak duality states that for any feasible solution x to the primal problem and any feasible solution y to the dual problem in the Diet Problem (Slides 2-8), the objective function value of the primal problem ($\sum_j c_j x_j$) is less than or equal to the objective function value of the dual problem ($\sum_i b_i y_i$). Explain why this inequality holds intuitively, considering the economic interpretations of the primal and dual problems.

   a. The primal problem minimizes cost, while the dual problem maximizes profit; therefore, cost is always less than profit.

   b. The primal problem's constraints are inequalities, while the dual problem's constraints are equalities, leading to a difference in objective function values.

   c. The primal problem focuses on food consumption, while the dual problem focuses on pill sales; these are distinct economic activities.

   d. The cost of obtaining nutrients through food is always less than or equal to the maximum profit from selling those same nutrients as pills because of market inefficiencies.

   e. The cost of a diet that meets nutritional requirements is always less than or equal to the maximum profit that can be made from selling pills that provide those same nutrients because the cost of production for pills is always less than the cost of food.

10. a. Consider the Diet Problem presented in Slides 2-8. The primal problem minimizes the cost of a diet while meeting nutritional requirements. The dual problem, introduced in Slides 7 and 8, maximizes the profit from selling nutrient pills. If the optimal solution to the primal problem results in a minimum cost of C^*, and the optimal solution to the dual problem results in a maximum profit of P^*, what relationship exists between C^* and P^* according to strong duality, and what is the economic interpretation of this relationship?

A. $C^* < P^*$; the minimum cost of the diet is always less than the maximum profit from selling nutrient pills.

B. $C^* > P^*$; the minimum cost of the diet is always greater than the maximum profit from selling nutrient pills.

C. $C^* = P^*$; the minimum cost of the diet equals the maximum profit from selling nutrient pills, reflecting the economic equivalence between the two problems.

D. $C^* \leq P^*$; the minimum cost of the diet is less than or equal to the maximum profit from selling nutrient pills, but the equality only holds under specific conditions.

E. There is no guaranteed relationship between C^* and P^*; their values depend on the specific parameters of the diet problem.

b. In the Diet Problem (Slides 2-8), the primal problem minimizes the cost of food while meeting nutritional requirements. The dual problem maximizes the profit from selling nutrient pills. Explain the economic interpretation of the dual variable y_i in the context of the dual problem's objective function, $\sum_i b_i y_i$.

A. y_i represents the cost of food i.

B. y_i represents the amount of nutrient i consumed.

C. y_i represents the profit from selling one unit of nutrient i in pill form.

D. y_i represents the minimum required intake of nutrient i.

E. y_i represents the amount of food i consumed.

2 Answers

1. In the Diet Problem (Slides 2-8), the primal problem minimizes the cost of food while meeting nutritional requirements. The dual problem, introduced in Slides 7 and 8, maximizes the profit from selling nutrient pills. Explain the economic interpretation of the dual variable y_i in the context of the dual problem's objective function, $\sum_i b_i y_i$.

   a. This is incorrect. y_i is not related to the amount of nutrient i consumed; that is represented by the primal variables x_j.

   b. This is incorrect. The cost of food j is represented by c_j, not y_i. y_i is associated with the price of a pill for nutrient i.

   c. The dual variable y_i represents the shadow price or marginal value of nutrient i. In the context of the dual problem, it represents the maximum price per unit of nutrient i that can be charged for a pill such that it remains more profitable to sell pills than to sell the original food. The objective function $\sum_i b_i y_i$ sums the profits from selling pills for each nutrient, considering the minimum required intake b_i for each nutrient.

   d. This is incorrect. The minimum amount of nutrient i required is represented by b_i, not y_i. y_i is the price of a pill for nutrient i.

   e. This is incorrect. The profit from selling one unit of food j is not directly represented by y_i. y_i is related to the price of a pill for nutrient i.

2. Consider the Diet Problem's primal formulation (Slide 5): minimize $\sum_j c_j x_j$ subject to $\sum_j a_{ij} x_j \geq b_i$ and $x_j \geq 0$. The dual problem (Slide 8) is formulated as: maximize $\sum_i b_i y_i$ subject to $\sum_i y_i a_{ij} \leq c_j$ and $y_i \geq 0$. Explain the economic interpretation of the dual constraint $\sum_i y_i a_{ij} \leq c_j$.

   a. This is incorrect. The constraint ensures that the pill cost is less than or equal to the food cost, not the other way around.

   b. The constraint $\sum_i y_i a_{ij} \leq c_j$ ensures that the total cost of pills needed to replace the nutrients in one unit of food j is less than or equal to the cost of one unit of food j. If this constraint were violated, it would be more profitable to buy the individual pills than to buy the food itself, which contradicts the economic logic of the dual problem.

   c. This describes a constraint in the primal problem, not the dual problem.

   d. This is incorrect. The constraint relates the cost of pills to the cost of food, not the profit from selling pills.

   e. This describes the non-negativity constraint on a_{ij} in the primal problem, which is not directly related to the dual constraint in question.

3. Weak duality states that for any feasible solution x to the primal problem and any feasible solution y to the dual problem, the objective function value of the primal problem is less than or equal to the objective function value of the dual problem. In the context of the Diet Problem (Slides 2-8), explain why this inequality holds intuitively.

   a. While the primal minimizes and the dual maximizes, the inequality holds because the dual provides an upper bound on the primal's optimal value, not simply because of the minimization and maximization nature.

   b. The use of food versus pills is a conceptual tool to illustrate duality; the inequality holds regardless of the specific items used.

c. Weak duality holds because the dual problem provides an upper bound on the optimal solution of the primal problem. The dual problem represents the maximum profit that could be obtained by selling pills that provide the same nutrients as the food in the primal problem. Since the primal problem aims to minimize the cost of obtaining the nutrients, the minimum cost will always be less than or equal to the maximum profit from selling the equivalent nutrients as pills.

d. Both primal and dual constraints can be inequalities; the inequality in weak duality arises from the relationship between the primal and dual objective functions.

e. The objective functions are related through the duality relationship, which is the basis for weak duality.

4. In the Diet Problem (Slides 2-8), strong duality states that if both the primal and dual problems have optimal solutions, then the optimal objective function values are equal. Explain the economic interpretation of this equality in the context of the problem.

   a. The equality holds only under the condition that both the primal and dual problems have optimal solutions; it's not a universal equality.

   b. Strong duality has a significant economic interpretation, reflecting an equilibrium between the cost of obtaining nutrients and the profit from selling them.

   c. Strong duality implies that if an optimal diet (minimum cost) is found, there exists a set of prices for nutrient pills such that the maximum profit from selling these pills exactly equals the minimum cost of the optimal diet. This reflects an economic equilibrium where the cost of obtaining nutrients through food is exactly balanced by the profit from selling the equivalent nutrients as pills.

   d. The pill prices and food prices are related but not necessarily equal; the equality in strong duality reflects an economic equilibrium, not a price equality.

   e. The minimum nutrient intake is a constraint in the problem; the equality in strong duality holds regardless of the specific value of the minimum intake.

5. Consider the primal Diet Problem (Slide 5): minimize $\sum_j c_j x_j$ subject to $\sum_j a_{ij} x_j \geq b_i$ and $x_j \geq 0$. Its dual (Slide 8) is: maximize $\sum_i b_i y_i$ subject to $\sum_i y_i a_{ij} \leq c_j$ and $y_i \geq 0$. If the primal problem represents minimizing the cost of a diet meeting nutritional requirements, what does the dual problem represent economically, and what is the economic interpretation of the dual variable y_i?

   a. While the dual problem involves selling nutrients, it's not about maximizing profit from selling individual nutrients but rather a complete set of nutrient pills that collectively meet the nutritional requirements.

   b. The dual problem is about maximizing profit, not minimizing production costs. The dual variables represent prices, not costs.

   c. The dual problem represents maximizing the profit from selling a complete set of nutrient pills, where each pill provides a specific nutrient. The dual variable y_i represents the maximum price that can be charged per unit of a pill containing nutrient i while remaining competitive with the cost of the original foods. This is because the dual constraint $\sum_i y_i a_{ij} \leq c_j$ ensures that the cost of pills to replace the nutrients in food j does not exceed the cost of food j itself.

   d. The dual problem does not directly address maximizing nutritional value; it focuses on maximizing profit from selling nutrient pills.

   e. The dual problem does not involve selling food; it involves selling nutrient pills as a substitute for food.

6. In the Diet Problem (Slides 2-8), the primal problem minimizes the cost of food while meeting nutritional requirements. The dual problem (Slides 7-8) maximizes the profit from selling nutrient pills. If the optimal solution to the primal problem is x^*, resulting in a minimum cost of $C^* = \sum_j c_j x_j^*$, and the optimal solution to the dual problem is y^*, resulting in a maximum profit of $P^* = \sum_i b_i y_i^*$, what relationship exists between C^* and P^* according to strong duality, and what is the economic interpretation of this relationship?

   a. This is incorrect. While it might seem intuitive that profit could exceed cost, strong duality establishes equality under optimal conditions.

   b. This is incorrect. Weak duality states that $C^* \leq P^*$, but strong duality refines this to equality under optimal conditions.

   c. According to strong duality, if both the primal and dual problems have optimal solutions, then their optimal objective function values are equal: $C^* = P^*$. Economically, this means that the minimum cost achievable by consuming foods to meet nutritional requirements is exactly equal to the maximum profit achievable by selling nutrient pills that satisfy those same requirements. This equality reflects the economic equilibrium between the cost of obtaining nutrients through food and the profit from providing them through pills.

   d. This is incorrect. While weak duality holds ($C^* \leq P^*$), strong duality states that $C^* = P^*$ when both primal and dual problems have optimal solutions.

   e. This is incorrect. Strong duality establishes a precise relationship between the optimal values of the primal and dual problems.

7. Consider the Diet Problem's primal formulation (Slide 5): minimize $\sum_j c_j x_j$ subject to $\sum_j a_{ij} x_j \geq b_i$ and $x_j \geq 0$. The dual problem (Slide 8) is: maximize $\sum_i b_i y_i$ subject to $\sum_i y_i a_{ij} \leq c_j$ and $y_i \geq 0$. Provide an economic interpretation of the dual constraint $\sum_i y_i a_{ij} \leq c_j$ in the context of the problem.

   a. This describes a primal constraint, not a dual constraint.

   b. The dual constraint $\sum_i y_i a_{ij} \leq c_j$ states that the total value of the nutrients in one unit of food j, calculated as the sum of the unit prices of the nutrients (y_i) multiplied by their respective amounts in food j (a_{ij}), must be less than or equal to the cost of one unit of food j (c_j). Economically, this means that the profit from selling the nutrients in food j as pills cannot exceed the cost of buying food j. If it did, it would be more profitable to buy the nutrients as pills rather than buying the food itself.

   c. This also describes a primal constraint.

   d. This describes the primal objective function.

   e. This describes a dual non-negativity constraint.

8. In the context of the Diet Problem (Slides 2-8), explain the economic interpretation of the dual variable y_i in the dual problem's objective function, $\sum_i b_i y_i$.

   a. This is the definition of x_j in the primal problem.

   b. This is the definition of b_i in both the primal and dual problems.

   c. In the dual problem of the Diet Problem, the objective function $\sum_i b_i y_i$ represents the total profit from selling nutrient pills. Therefore, y_i represents the unit price (or shadow price) of a pill that provides one unit of nutrient i. The dual problem seeks to maximize this profit, subject to constraints that ensure the total value of nutrients in each food does not exceed its cost.

   d. This is the definition of a_{ij} in both the primal and dual problems.

e. This is the definition of c_j in the primal problem.

9. Weak duality states that for any feasible solution x to the primal problem and any feasible solution y to the dual problem in the Diet Problem (Slides 2-8), the objective function value of the primal problem ($\sum_j c_j x_j$) is less than or equal to the objective function value of the dual problem ($\sum_i b_i y_i$). Explain why this inequality holds intuitively, considering the economic interpretations of the primal and dual problems.

   a. While the primal minimizes and the dual maximizes, the inequality is not always strict. Strong duality shows equality under optimal conditions.

   b. Both primal and dual constraints can be inequalities. The inequality arises from the economic relationship between the cost of food and the value of its nutrients.

   c. While the activities are different, the inequality arises from the economic relationship between the cost of food and the value of its nutrients, not simply the difference in activities.

   d. Weak duality reflects the fact that the cost of obtaining nutrients through food will always be less than or equal to the maximum profit that can be made from selling those same nutrients as pills. This is because the cost of food includes not only the cost of the nutrients but also the cost of other components and processing. The dual problem only considers the value of the nutrients themselves, ignoring other factors. Therefore, the maximum profit from selling pills can be greater than or equal to the minimum cost of a diet, but never less.

   e. This is an overly simplistic and potentially incorrect assumption about production costs. The inequality holds due to the inclusion of non-nutrient components and processing costs in food production.

10. a. Consider the Diet Problem presented in Slides 2-8. The primal problem minimizes the cost of a diet while meeting nutritional requirements. The dual problem, introduced in Slides 7 and 8, maximizes the profit from selling nutrient pills. If the optimal solution to the primal problem results in a minimum cost of C^*, and the optimal solution to the dual problem results in a maximum profit of P^*, what relationship exists between C^* and P^* according to strong duality, and what is the economic interpretation of this relationship?

      A. This is incorrect. While it might seem intuitive that the cost of a diet could be less than the profit from selling pills, strong duality establishes an equality under optimal conditions.

      B. This is incorrect. Weak duality states that $C^* \leq P^*$, but strong duality refines this to an equality under optimal conditions.

      C. Strong duality states that if both the primal and dual problems have optimal solutions, then their optimal objective function values are equal. In the context of the Diet Problem, this means the minimum cost of the diet (C^*) equals the maximum profit from selling nutrient pills (P^*). This reflects an economic equivalence: the minimum cost achievable by consuming food is the same as the maximum profit achievable by selling the equivalent nutrients in pill form. This equality only holds if both the primal and dual problems are feasible and bounded.

      D. This is partially correct in stating weak duality, but strong duality implies equality under the conditions stated in option C.

      E. This is incorrect. Strong duality guarantees a specific relationship between the optimal values of the primal and dual problems.

    b. In the Diet Problem (Slides 2-8), the primal problem minimizes the cost of food while meeting nutritional requirements. The dual problem maximizes the profit from selling nutrient pills. Explain the economic interpretation of the dual variable y_i in the context of the dual problem's objective function, $\sum_i b_i y_i$.

A. This is incorrect. The primal problem deals with the cost of food, while the dual problem focuses on the profit from selling pills.

B. This is incorrect. The amount of nutrient i consumed is represented by a variable in the primal problem.

C. In the dual problem, the objective function $\sum_i b_i y_i$ represents the total profit from selling nutrient pills. Each term $b_i y_i$ represents the profit obtained from selling the minimum required amount (b_i) of nutrient i at a price of y_i per unit. Therefore, y_i represents the unit price or profit obtained from selling one unit of nutrient i in pill form.

D. This is incorrect. The minimum required intake of nutrient i is represented by b_i in both the primal and dual problems.

E. This is incorrect. The amount of food i consumed is a variable in the primal problem, not the dual.